

GRÉGOIRE SZYMANSKI

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EDUCATION

PhD in Mathematics (supervised by M. Rosenbaum and M. Hoffmann) <i>École Polytechnique, CMAP</i>	Dec. 2024 <i>Palaiseau, France</i>
Diplôme de l'ENS in Mathematics <i>École Normale Supérieure</i>	Sept. 2021 <i>Paris, France</i>
Master El Karoui – Master of Science in Quantitative Finance <i>Sorbonne Universités</i>	Sept. 2020 <i>Paris, France</i>
Bachelor's degree in Mathematics and Computer Science <i>École Normale Supérieure</i>	Sept. 2018 <i>Paris, France</i>

ACADEMIC POSITIONS

Postdoctoral researcher (supervised by M. Podolskij) <i>University of Luxembourg, Department of Mathematics</i>	Sept. 2024 – Sept. 2027 <i>Esch-sur-Alzette, Luxembourg</i>
PhD student (supervised by M. Rosenbaum and M. Hoffmann) <i>École Polytechnique, CMAP</i>	Sept. 2021 – Sept. 2024 <i>Palaiseau, France</i>
Undergraduate Research Assistant <i>École Polytechnique, CMAP</i>	Feb. 2021 – Sept. 2021 <i>Palaiseau, France</i>
Research Internship <i>University of Helsinki</i>	Feb. 2019 – June 2019 <i>Helsinki, Finland</i>
Research Internship <i>Paris Diderot</i>	June 2017 – July 2017 <i>Paris, France</i>

PUBLICATIONS AND PREPRINTS

- Grégoire Szymanski (2024): Optimal estimation of the rough Hurst parameter in additive noise. **Stochastic Processes and their Applications.**
- Carsten Chong, Marc Hoffmann, Yanghui Liu, Mathieu Rosenbaum and Grégoire Szymanski (2023): Statistical inference for rough volatility: Minimax Theory. **The Annals of Statistics.**
- Carsten Chong, Marc Hoffmann, Yanghui Liu, Mathieu Rosenbaum and Grégoire Szymanski (2023): Statistical inference for rough volatility: Central limit theorems. **The Annals of Applied Probability.**
- Bruno Durin, Mathieu Rosenbaum and Grégoire Szymanski (2023): The two square root laws of market impact and the role of sophisticated market participants. **To appear in The Annals of Applied Probability.**
- Grégoire Szymanski and Tetsuya Takabatake (2023): Asymptotic Efficiency for Fractional Brownian Motion with general noise. **Submitted.**
- Raphael Maillet and Grégoire Szymanski (2024): Estimation of the invariant measure of a multidimensional diffusion from noisy observations. **Submitted.**
- Youssef Ouazzani Chahdi, Mathieu Rosenbaum and Grégoire Szymanski (2026): Passive Market Impact: A Point Process Approach. **To appear in Finance and Stochastics.**
- Grégoire Szymanski and Wei Xu (2025): Mean-Field Limits for Nearly Unstable Hawkes Processes. **Submitted.**

9. Youssef Ouazzani Chahdi, Nathan De Carvalho and Grégoire Szymanski (2026): Trading with market resistance and concave price impact. **Submitted.**
10. Florian Bourgey, Mathieu Rosenbaum, Patrick Noble, Grégoire Szymanski and Ingi Petursson (2026): The Quadratic Rough Heston+ Model for Short-Dated Options. **Submitted.**
11. Johannes Muhle-Karbe, Youssef Ouazzani Chahdi, Mathieu Rosenbaum and Grégoire Szymanski (2026): A unified theory of order flow, market impact, and volatility. **Submitted.**

RESEARCH VISITS

University of Hiroshima <i>Invited by Tetsuya Takabatake</i>	Aug – Sept. 2023 <i>Hiroshima, Japan</i>
Hong Kong University of Science and Technology <i>Invited by Carsten Chong</i>	Sept. 2023 <i>Hong Kong, China</i>
Beijing Institute of Technology <i>Invited by Wei Xu</i>	Sept. 2023 <i>Liangxiang, China</i>

ACADEMIC SERVICE

Reviewer <i>The Annals of Applied Probability, Quantitative Finance, SPA, Electronic Journal of Statistics, Finance and Stochastics, Decisions in Economics and Finance, Bernoulli, and SIAM Journal on Financial Mathematics</i>	Since 2022
Co-Organizer of the FinML Workshop - Mathematical Insights from Markets, Control, and Learning <i>with Lorenzo Croissant and Antonio Ocello</i>	2024 <i>Aussois, France</i>
Co-Organizer of the AMART Seminar - Applied Mathematics and Related Topics at the University of Luxembourg <i>with Lorenzo Cristofaro and Francesca Pistolato</i>	2025 – 2026 <i>Esch-sur-Alzette, Luxembourg</i>

TEACHING EXPERIENCE

Mathematics (High School, TALENS section) Lecture and Exercises, ENS	2018 – 2019
How to Write Mathematics (Bachelor 1st year) Lecture and Exercises, École Polytechnique	2021 – 2022
Processus stochastiques (M2 Mathématiques pour les Sciences du Vivant) Exercises, École Polytechnique	2021 – 2022
Probabilistic Models in Finance (Master of Mathematics) Lecture and Exercises, University of Luxembourg	Since 2024
Logiciels mathématiques (Master of Mathematics) Lecture and Exercises, University of Luxembourg	2025
Continuous Time Models in Mathematical Finance (Master of Mathematics) Exercises, University of Luxembourg	Since 2025
Bayesian Statistics (Master of Mathematics) Exercises, University of Luxembourg	Since 2025

STUDENT SUPERVISION AND MENTORING

Master's students

Ilya Ilchenko, Attilio Pitelli, Anubhav Roy, Gaurav Yengera

Bachelor's students

Karel Moryoussef, Adan Fhima, Catia Alves Pires

PROFESSIONAL EXPERIENCE

Jump Trading

ILB Chair "Artificial Intelligence and Quantitative Methods for Finance"

Since 2025

Paris, France

Welwing Capital

Consultant

Since 2022

Paris, France

Melanion Capital

Consultant

2025

Paris, France

BNP Paribas

Quantitative analyst intern

June 2020 – Dec. 2020

London, UK

EDF

Partnership with ENS

Sept. 2019 – Dec. 2019

Paris, France

SEMINARS AND CONFERENCE PRESENTATIONS

CEREMADE Winter School, Domaine de la Tour, Saint Pierre Canivet, France

Feb. 27 – Mar. 4, 2022

DynStoch 2022, Institut Henri Poincaré, Paris, France

June 29 – July 1, 2022

StatMathAppli 2022, La Villa Clythia, Fréjus, France

Aug. 29 – Sept. 2, 2022

Séminaire des jeunes chercheurs du Ceremade, Université Paris Dauphine, Paris, France

Oct. 13, 2022

Groupe de travail des thésards du LPSM, Jussieu, Paris, France

Nov. 7, 2022

Séminaire des doctorants du CMAP, École Polytechnique, Palaiseau, France

Feb. 22, 2023

Groupe de Travail du CERMICS, École des Ponts, Palaiseau, France

May 15, 2023

Workshop on modeling with Hawkes processes: Theory, Statistics and Applications, Institute of Mathematics of the Polish Academy of Sciences, Warsaw, Poland

May 29 – June 2, 2023

Summer School on Big Data and Finance, Centre Paul Langevin, Aussois, France

June 12 – June 16, 2023

ICIAM 2023, Waseda University, Tokyo, Japan

Aug. 21 – Aug. 25, 2023

Seminar, Beijing Institute of Technology, Beijing, China

Sept. 21, 2023

Séminaire des doctorants du CMAP, École Polytechnique, Palaiseau, France

Nov. 22, 2023

Séminaire des doctorants du CEREMADE, Université Paris Dauphine, Paris, France

Nov. 23, 2023

EFFI France-Japan seminar, Le Mans Université, Le Mans, France

Dec. 5, 2023

Matinée recherche CFM , Capital Fund Management, Paris, France	Mar. 12, 2024
SMSA 2024 , TU Delft, Delft, Netherlands	Mar. 13 – Mar. 15, 2024
Third-year PhD Students’ Day , Université Paris Dauphine, Paris, France	Mar. 26, 2024
FinML Workshop , Centre Paul Langevin, Aussois, France	Sept. 23 – Sept. 27, 2024
Prob&Stat Seminar , University of Luxembourg, Esch-sur-Alzette, Luxembourg	Oct. 17, 2024
Luxembourg Workshop in Stochastic Analysis , University of Luxembourg, Esch-sur-Alzette, Luxembourg	Feb. 12, 2025
17th Annual SoFiE Conference , ESSEC Campus of Cergy, Cergy, France	June 9 – June 12, 2025
SIAM Conference on Financial Mathematics and Engineering , Miami, FL, USA	July 15 – July 18, 2025
Mathematical Finance Seminar , Columbia University, New York, NY, USA	Sept. 4, 2025
Young Researchers in Stochastic Analysis and Stochastic Geometric Analysis , Bernoulli Center, EPFL, Lausanne, Switzerland	Sept. 23 – Sept. 26, 2025
IMS International Conference on Statistics and Data Science 2025 , Sevilla, Spain	Dec. 15 – Dec. 18, 2025
2nd Dolomites Winter School: Mean-Field Systems in Finance, Neurosciences, and AI , Centro Congressi - Folgarida and Madonna di Campiglio, Folgarida, Italy	Jan. 11 – Jan. 16, 2026
KCL Mathematical Finance Seminar , King’s College London, London, UK	Mar. 2, 2026
Séminaire sur les processus de Hawkes du LPSM , Sorbonne Université, Paris, France	Mar. 16, 2026
XXVII Workshop on Quantitative Finance , University of Bergamo, Bergamo, Italy	Mar. 30 – Apr. 1, 2026
Statistical Inference for Particle Systems , Institut Henri Poincaré and Université Évry Paris-Saclay, Paris and Évry, France	Apr. 7 – Apr. 8, 2026
13th Days of Econometrics for Finance , Toulouse School of Economics, Toulouse, France	May 22 – May 23, 2026
Financial Econometrics Conference , Mohammed V University in Rabat, Rabat, Morocco	Apr. 28 – Apr. 30, 2026
30th Nordic Conference in Mathematical Statistics , University of Helsinki, Helsinki, Finland	June 1 – June 4, 2026
DynStoch 2026 , Chalmers University of Technology and the University of Gothenburg, Gothenburg, Sweden	June 15 – June 17, 2026
13th World Congress of the Bachelier Finance Society , University of Bologna, Bologna, Italy	June 29 – July 3, 2026
25th European Young Statisticians Meetings , Vilnius, Lithuania	July 7 – July 10, 2026
Vienna-Copenhagen Conference on Financial Econometrics , University of Vienna, Vienna, Austria	Aug. 13 – Aug 15, 2026